

The Wrong Unit of Uncertainty: Simultaneous Inference for Repeated Cross-National Surveys

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Abstract

Repeated cross-national surveys are used to make claims about whole trajectories of national response distributions, while uncertainty is computed for single wave contrasts; and when country trajectories calibrate cross-country prediction, the calibration objects are themselves estimates from complex samples. We develop simultaneous inference that attaches uncertainty to both units. Within a country, one design-based band certifies a partially ordered family of trajectory claims — from a single wave pair to persistent, distribution-wide decline — with no further correction. Across countries, a clustered conformal band gives finite-sample coverage for survey-estimated trajectories; we prove that correcting toward the latent trajectory is unidentified without design-noise information, and characterize when the correction is unnecessary, unlearnable, or feasible — infeasible at cross-national scale, active at the small-area unit of the same survey. In the European Social Survey a marginal reading flags twenty of thirty countries; six certify net decline over 2018–2024, and over 2002–2024 none certifies a persistent slide — a claim only crisis-scale change clears — while closed testing certifies at least six true span decliners among thirty-three. The same discipline cuts the certified

WVS/EVS deconsolidation set severalfold, toward post-communist and Arab-Spring states.

1 Introduction

Comparative politics increasingly treats a national public as a *distribution* rather than a mean. Trust in parliament in a country and year is a cumulative distribution function over a response scale, and repeated cross-national surveys—the European Social Survey (ESS; Kaminska and Lynn, 2017), the World Values Survey (WVS), the AmericasBarometer (Cohen et al., 2021)—generate claims about how those distributions evolve across waves. A second use follows: *transporting* the trajectories, predicting one country’s from the others’, for which conformal prediction promises finite-sample, distribution-free coverage (Vovk et al., 2005; Lei et al., 2018).

Two wrong units. One principle organizes this paper: *the stochastic unit of a guarantee must coincide with the unit of the scientific claim it certifies*. Claims from repeated surveys violate it twice. A common inferential move builds the *claim* — “democratic trust is declining” — from a single wave-pair contrast and narrates it as a country-level trend (Foa and Mounk, 2016, 2017; Valgarðsson et al., 2025). The second, subtler wrong unit is the calibration: the objects conformal transport calibrates on are themselves *estimated* from complex probability samples, and existing constructions plug them in as if exact (Dunn et al., 2023; Lee et al., 2026).

The two errors are distinct. Writing $F_{c,r}(t)$ for country c ’s latent CDF at threshold t in wave r and $\tilde{F}_{c,r}(t)$ for its complex-survey estimate,

$$\tilde{F}_{c,r}(t) - \mu_r(t) = \underbrace{(F_{c,r}(t) - \mu_r(t))}_{\text{transport error}} + \underbrace{(\tilde{F}_{c,r}(t) - F_{c,r}(t))}_{\text{design error}}, \quad (1)$$

where μ_r is the transport center. A band calibrated on the observed $\tilde{F}$ scores treats the sum

as the transport error alone: when the design error is non-negligible the band is too wide, yet any correction subtracting the survey noise can make it too narrow, being itself estimated from few countries.

Contributions. We make four. *First, a procedure for the claim unit:* a design-based simultaneous band over a country’s whole trajectory, from which a partially ordered family of claims — from a single wave pair up to persistent, distribution-wide decline — is read off one coverage event (Section 3.1). *Second, an impossibility theorem for the calibration unit.* Without information on the design-noise law the latent transport-error law is non-identified, and no band built on the observed curves can uniformly improve on the plug-in conformal band beyond its discreteness slack while remaining valid (Theorem 1). Estimated calibration is not a nuisance to correct away; it is an inferential boundary. *Third, the feasibility boundary of the correction, and a selector that respects it.* Even with the design-noise information supplied, the correction’s reliability diagnostic obeys an algorithmic floor $D \geq \sqrt{2/(K-1)}$, so the deployed correction requires $K \geq 94$ exchangeable populations — a bound the two largest surveys fail in opposite ways, while one unit down, on regions’ departures from their national distributions ($K \approx 230$ – 290), the same frozen selector does activate (Section 6). The deployed band is exact at any K and selection among its branches costs no validity (Theorems 3 and 5). (The clustered band is machinery shared with a companion paper on a non-survey domain, Park, 2026; new here is what the survey layer forces — the impossibility, the floor, the safe selector, and the claim-family deployment.) *Fourth, two reanalyses where the procedures change the substantive picture* (Section 7). For European trust the family certifies net decline in six of thirty countries (2018–2024), not the twenty flagged marginally; over the full 2002–2024 record no country certifies persistence, while twenty-three of thirty-three certify both declines and recoveries and eight certify span erosion, all read off one band per country. Globally, applying a trajectory-persistence criterion to the Foa–Mounk “deconsolidation” battery on the WVS/EVS (1981–2022) cuts the wave-pair certified set by

2.6–6.5 $\times$, and the surviving thirteen-country *certified core* concentrates in post-communist and Arab-Spring-aftermath states.

2 Setup and the two targets

Objects. We observe K source countries, indexed $c = 1, \dots, K$, treated as exchangeable units (the exchangeable object is the country, not the respondent). Each country has a latent *response-distribution trajectory* $F_{c,r}(t)$, a CDF over a fixed grid of thresholds $t \in \{t_1, \dots, t_T\}$ and survey rounds r . We say *response distribution* deliberately: guarantees attach to recorded answers, and reading a certified shift as an *attitude* shift additionally requires measurement invariance across waves (Section 8); one invariance comes free, since every claim below survives monotone recoding of the ordinal scale. We do not see $F_{c,r}$; we see a complex-survey estimate $\tilde{F}_{c,r}(t) = F_{c,r}(t) + S_{c,r}(t)$, where $S_{c,r}$ is design (sampling) error with a design variance $v_{c,r}(t)^2$ that a stratified primary-sampling-unit (PSU) / replicate-weight bootstrap can estimate. The transport score of a country is its worst-case standardized deviation from the transport center,

$$R_c = \max_{r,t} \frac{|F_{c,r}(t) - \mu_r(t)|}{\sigma(t)}, \quad \tilde{R}_c = R_c + \xi_c,$$

where $\tilde{R}_c$ is the same quantity computed from the observed $\tilde{F}$ and ξ_c is the induced design contamination (a working approximation, since R is a supremum; the impossibility below is stated at the curve scale and does not use it). The governing scalar is the design-to-total SD ratio

$$\rho = \frac{\text{SD}(\xi)}{\text{SD}(\tilde{R})} = \sqrt{\frac{\overline{v^2}}{s_R^2 + \overline{v^2}}} \in [0, 1),$$

$\rho \rightarrow 0$ is the design-noise-negligible regime, $\rho \rightarrow 1$ the noise-dominated one.

Target and validity. We want a simultaneous band covering a new country's *latent* trajectory, $\Pr(F_{\text{new},r}(t) \in \hat{B}(r, t) \forall r, t) \geq 1 - \alpha$. Ordinary clustered conformal on the observed

Procedure	Target	Key assumption	Guarantee
Clustered conformal band (anchor)	survey-estimated trajectory	country exchangeability	finite-sample, every K , under symmetric centers (Thm 3)
Same band, latent reading	latent trajectory	+ design-noise tail and anti-concentration	$1-\alpha$ -gap, gap shrinking with ρ (Thm 2b)
Design-aware correction	latent trajectory	design law (A1)–(A3), $K \geq 94$	$1-\alpha-\varepsilon_{K,B}$; deployed level a frozen calibrated bound (Thm 4)
Certification band (§3.1)	within-country contrast surface	survey-design bootstrap	asymptotic simultaneous $1-\alpha$

Table 1: Which procedure guarantees what. The conservative envelope (omitted) shares the anchor’s row, containing it by construction; the theory’s finite-sample language does not transfer to the certification band.

scores is exact for the *survey-estimate* target $\tilde{F}_{\text{new}}$, needing only exchangeability; its latent coverage differs by an explicit gap vanishing with the design-noise scale (Theorem 2(b); symmetry alone is not enough), its width is inflated by design noise, and correcting that is a deconvolution. Table 1 collects the paper’s procedure–target–guarantee triples.

What country exchangeability assumes. The transport band’s identifying assumption is a superpopulation postulate: participating countries’ trajectory laws are symmetric draws, over a set that is not sampled and enters rounds non-randomly — a strong idealization, tempered three ways. The headline certifications of Section 7 do not use it; under approximate exchangeability the Barber et al. (2023) bound controls the gap; and a leave-one-region-out audit finds near-nominal held-out coverage except for the post-communist East, a quantified scope condition on transport *into* that bloc (Supplementary Material).

The impossibility. The central obstruction is that the deconvolution is not identified from the contaminated curves alone. The result is stated at the curve level of (1), where the additive model *is* the survey sampling structure; the design noise may be arbitrarily heteroskedastic across thresholds.

Theorem 1 (Non-identification and necessity of design information). *Let the observed deviation curves be $\tilde{E}_c = E_c + S_c$ as in (1), the latent curves exchangeable with law $\mathcal{L}_E$, the design-noise curves independent of them with law $\mathcal{S}$ in an admissible class that contains $S \equiv 0$ and is closed under one further nondegenerate symmetric convolution. (i) $\mathcal{L}_E$ is not identified from the observed law $\mathcal{L}_E * \mathcal{S}$. (ii) Any band whose radius is measurable in the observed curves alone and is valid for the latent target over the whole admissible class cannot uniformly improve on the plug-in conformal band: on the member $S \equiv 0$, which it cannot distinguish from the contaminated configurations, the plug-in radius is already the tightest valid one up to the conformal discreteness slack, which randomized smoothing exhausts. Uniform improvement requires shrinking the class of noise laws.*

The reading is constructive (proof in the Supplementary Material, Section S1): a band both valid and narrower than the plug-in must supply the design-noise law — exactly what Section 3’s design bootstrap does.

Remark 1 (Beyond surveys). Nothing here is survey-specific: the result holds whenever conformal calibration uses estimated objects — model-based small-area outputs, learned or imputed features, meta-analytic effects. Design/measurement information is the identifying ingredient, and its reliability at finite K is itself bounded (Proposition 2); confirmed outside surveys on a small-area calibration (Supplementary Material).

Relation to existing work. Conformal prediction has met complex surveys for unit-level prediction (Wieczorek, 2023), and cluster-as-unit constructions transport guarantees to an unseen cluster while treating its data as cleanly sampled (Dunn et al., 2023; Lee et al., 2026); our calibration objects are instead estimated under complex designs (Binder, 1983; Lumley, 2010). Noise-aware conformal methods correct under a *known* noise model (Einbinder et al., 2024; Sesia et al., 2025; Singer et al., 2025); Theorem 1 says that premise fails for design noise — a deconvolution problem (Fan, 1991) — and Proposition 2 that even a supplied law faces a finite- K floor.

3 Simultaneous inference for both units

The two wrong units get two procedures: within-country certification of a whole claim family, and cross-country transport with a correction that activates only where it is provably reliable.

3.1 Within-country certification: one band, a family of claims

The claims of interest differ in their unit. A *pairwise* claim asks whether a given adjacent pair of waves shows a distributional shift; *any-pair* whether some adjacent pair does; *net* whether the first-to-last span declined; *persistent country-wide* whether the whole distribution declined simultaneously across the trajectory; *Bonferroni* further controls the number of countries examined. These claims form a partial order, not a chain: persistent implies both net and any-pair, but net and any-pair are incomparable — a dip that recovers certifies an adjacent pair with no net decline, and a slow accumulation can certify a long span while no single adjacent pair clears the critical value; both patterns occur in the ESS record (Section 7). The claims are not separate tests: each claim, every sub-span, and the reverse direction is a function of one object, the country’s surface of wave-pair contrasts, so all are certified on one coverage event. Reading one claim as another is the over-detection this paper is about.

Proposition 1 (Claim-family transfer). *Let $\widehat{B}$ be a simultaneous $1 - \alpha$ band for a country’s contrast surface $\theta = \{\theta_{a,b}(t)\}$ over every ordered pair of observed rounds, and let $\{A_j\}$ be any family of claims, each a set of surfaces. Define the certified subfamily $J = \{j : \widehat{B} \subseteq A_j\}$. Then $\Pr(\theta \in A_j \text{ for every } j \in J) \geq 1 - \alpha$, with no dependence on $|J|$: on $\{\theta \in \widehat{B}\}$, set inclusion delivers every certified claim at once.*

The proof is the line just given; reading each deterministic consequence off one simultaneous region is the post-hoc logic of Scheffé (1953), closed testing (Marcus et al., 1976), and the selection-free bounds of Goeman and Solari (2011); Katsevich and Ramdas (2020). The contribution is its deployment: trend claims in this literature are made claim by claim,

and one studentized sup- t band over the whole contrast family removes the corrections a per-claim construction needs — no Bonferroni across a country’s pairs, no separate budget for certified recoveries, no endpoint-selection cost — at the price of a wider critical value, which we report. The persistent object is covered directly by one functional band (Diquigiovanni et al., 2022), a distinction of estimand rather than error rate (Gelman et al., 2012; Benjamini and Hochberg, 1995). The band itself is a design-based simultaneous sup- t band on wave-pair CDF differences from the stratified-PSU bootstrap (Rao and Wu, 1988; Wolter, 2007): an asymptotic guarantee in which exchangeability plays no role, and to which the theory sections’ finite-sample language does not transfer.

3.2 Cross-country transport and the safe adaptive band

Theorem 1 says the design bootstrap is the identifying input; the transport method turns it into a band by choosing, target-blind, among three constructions.

Three branches. For calibration error curves $\{\tilde{E}_c\}_{c=1}^K$ over threshold grid t , with per-country design SD $\hat{v}_c(t)$ from the stratified-PSU bootstrap: (i) *The clustered conformal band* takes the conformal quantile of the *unstudentized* scores $\max_t |\tilde{E}_c(t)|$: exact for the survey-estimate target at any K under Theorem 3’s symmetric centers, with the latent gap of Theorem 2(b). (ii) *Deconvolution* studentizes by the finite- K -guarded target scale $\hat{s}_T^2(t) = \max(s_{\text{plug}}^2(t) - [\hat{v}^2(t) - z \text{ SE}]_+, \text{ floor})$, which subtracts a *lower* confidence bound on the design variance so the correction cannot over-shrink; its width scales as $\sqrt{1 - \rho^2}$ relative to the clustered band. (iii) *Conservative* inflates each unstudentized score by the design SD, $\max_t (|\tilde{E}_c(t)| + z \hat{v}_c(t))$, a score that dominates the clustered band’s score pointwise, so the conservative band *contains* the clustered band by construction.

Safe selector (Algorithm 1). A naive selector asks *can we deconvolve* and *do we need to*; the safe selector adds the missing third question, *can we do it safely at this K ?* Deconvolution is used only when four gates pass, all computed from the calibration set alone: *need*

Algorithm 1: Nominal-safe adaptive design-aware band

Input: calibration curves $\{\tilde{R}_c\}$ (unit = country), design replicates $\{\hat{v}_c\}$, target curve $\hat{F}_{\text{new}}$, level α ; frozen gate constants and the K -only budget rule $\alpha_{\text{dec}}(K)$ set $(\alpha_{\text{anchor}}, \alpha_{\text{dec}})$ by the frozen budget rule (full α to the anchors when gate B is infeasible at this K);
compute diagnostics $s, \hat{s}_T, \hat{\rho}_{\text{LCB}}, D, \hat{\delta}_{\text{UCB}}(D)$; the nested unstudentized anchor radii $r_{\text{clu}}, r_{\text{con}}$ at level α_{anchor} ; and r_{dec} at level α_{dec} ;
if $\hat{\rho}_{\text{LCB}} \leq \rho_0$ **then**
 | branch $\leftarrow$ clustered;
else if $\hat{\delta}_{\text{UCB}}(D) \leq \delta_{\text{max}}$ **and** *stable* **and** $\widehat{\text{Gain}} - \Delta_g > g_{\text{min}}$ **then**
 | branch $\leftarrow$ deconvolution;
else
 | branch $\leftarrow$ conservative;
if branch $\neq$ deconvolution **and** the center is leave-one-out (the unsurveyed-target default) **then**
 | $r_{\text{branch}} \leftarrow \frac{K}{K-1} r_{\text{branch}}$; // exactness for the deployed center (Section 4)
return $\hat{F}_{\text{new}} \pm r_{\text{branch}}$, branch, and diagnostics;

(A: $\hat{\rho}_{\text{LCB}} > \rho_0$, a conservative lower confidence bound on ρ), *reliability* (B: $\hat{\delta}_{\text{UCB}}(D) \leq \delta_{\text{max}}$, a frozen upper confidence bound on the finite- K coverage remainder through the diagnostic $D = \max_t \text{SE}(\hat{s}_T^2)/\hat{s}_T^2$), *efficiency* (C: a certified width gain $\widehat{\text{Gain}}$ over the conservative band clearing a margined threshold), and *stability* (D); otherwise the band is an anchor, as Algorithm 1 routes. Every gate is target-blind, so by Theorem 5’s nested-band argument validity survives even adversarial selection between the anchor branches; the non-nested deconvolution branch is charged its own miss budget, and only in the $K \geq 94$ regime where its gate can open at all (Section 4). All constants are frozen before validation (Supplementary Material); when deconvolution is used the reported level is the observable $1 - \alpha - \hat{\delta}_{\text{UCB}}(D)$, a frozen *calibrated* bound (the theorem-level guarantee is Theorem 4’s $\varepsilon_{K,B}$). The method ships as one call, `dapcb(cal_errors, v_cal, center, alpha)`, returning the band, the selected branch and why, the coverage level, and the target that level refers to.

4 Theory

Theorem 3 anchors the validity architecture and Theorem 5 carries the guarantee through the selector; Theorems 2 and 4 price what the deployed band does not get for free — the latent rather than estimated trajectory, an estimated rather than known noise law. Proofs are in the Supplementary Material, Section S1; each theorem is paired with an automated contract test checking its computational implications.

Theorem 2 (Oracle validity). *Suppose the countries are exchangeable and the design law is known. (a) The clustered band is finite-sample valid for the survey-estimate target, $\Pr(\tilde{F}_{new} \in \hat{B}) \geq 1 - \alpha$, using exchangeability only. (b) For the latent target the same band satisfies, for every $u > 0$, $\Pr(F_{new} \in \hat{B}) \geq 1 - \alpha - \Pr(\xi < -u) - \Pr(\tilde{R} \in (\hat{q} - u, \hat{q}])$. (c) The deconvolution band with the true target scale is valid for the latent target, and finite-sample exact under the scale-family assumption (A1), with width a factor $\sqrt{1 - \rho^2} + o(1)$ of the clustered band.*

The gap in (b) is a noise-tail plus a local anti-concentration term, vanishing with the design-noise scale at rate $\rho^{2/3}$ under variance-only tails and ρ up to logarithms under sub-Gaussian noise — but not by symmetry (a bimodal counterexample, Supplementary Material): hence we estimate and report ρ rather than assume it away.

Theorem 3 (Exact finite- K validity of the deployed base band). *Let the base band use the unstudentized cluster score $R_c = \max_t |E_c(t)|$, with a center satisfying the symmetric-construction condition (the Supplementary Material states the four admissible cases). If the countries are exchangeable, then for every K the band $\hat{B} = \text{center} \pm \hat{q}$, with $\hat{q}$ the $\lceil (1 - \alpha)(K+1) \rceil$ -th order statistic of $\{R_c\}$, satisfies $\Pr(\tilde{F}_{new} \in \hat{B}) \geq \lceil (1 - \alpha)(K+1) \rceil / (K+1) \geq 1 - \alpha$, distribution-free.*

The guarantee is exact for the survey-estimate trajectory T1; the latent trajectory T2 adds the gap of Theorem 2(b), bounded through the reported ρ . Exactness at every K is bought by excluding two asymmetries — in-sample modulation and grand-mean centers

containing their own unit — at a width cost of at most 5% on the near-homoskedastic real curves.

Theorem 4 (Estimated-law validity). *Assume a common scale family indexed by the design variance (A1), anti-concentration of the sup-score law near its upper quantile (A2), and bounded curves with an unbiased size- B design bootstrap (A3); the appendix states all three precisely. Then (i) with oracle scales the deconvolution band is finite-sample exact for the latent target at every K , and (ii) with estimated scales it satisfies $\Pr(F_{new} \in \widehat{B}) \geq 1 - \alpha - \varepsilon_{K,B}$ with $\varepsilon_{K,B} = O(\sqrt{\log(KT)/\min(K, B)})$.*

The remainder’s direction is fixed by construction: the finite- K -guarded scale $\widehat{s}_T^2$ of Section 3 subtracts a lower confidence bound on the design variance, so outside an α_2 -probability event the scale error widens the band. Some shape restriction is not removable (the counterexample above); (A1) is one sufficient choice. The miss budget is a frozen function of K alone: below the reliability floor ($K < 94$, the regime in which the gate cannot open at all, Proposition 2, covering every certification sample here) the anchors keep the full α and the deconvolution budget is zero; at $K \geq 94$ a frozen K -only rule reserves α_{dec} (Supplementary Material).

Theorem 5 (Safe-adaptive finite- K validity). *Deploy Algorithm 1 with the unstudentized clustered and conservative branches, scores $\max_t |E_c(t)|$ and $\max_t (|E_c(t)| + z\widehat{v}_c(t))$, at level α_{anchor} , and the deconvolution branch at α_{dec} . Then, finite-sample and with no condition on the selection rule,*

$$\Pr(\widetilde{F}_{new,r}(t) \in \widehat{B}) \geq 1 - \alpha \quad (K < 94), \quad \Pr(F_{new,r}(t) \in \widehat{B}) \geq 1 - \alpha - \varepsilon_{K,B} \quad (K \geq 94).$$

The clauses name different targets because they rest on different assumptions: exchangeability alone delivers T1 exactly, with T2 adding Theorem 2(b)’s gap, while (A1)–(A3) deliver the latent target directly, by an anchor-domination lemma, with the remainder $\varepsilon_{K,B}$ charged to the deconvolution branch alone — the same-order term its leave-one-out centering adds

is absorbed into $\varepsilon_{K,B}$ (the Supplementary Material extends Theorem 4' to that construction). The proof uses no conditional-on-selection step: the conservative score dominates the clustered band's score pointwise, selection between nested bands anchored at an exact band is validity-free — nested-conformal machinery (Gupta et al., 2022; Yang and Kuchibhotla, 2024), with the contribution of engineering the branches so that it applies — and a union bound charges the one non-nested branch its own budget.

Two scope notes. At $K \geq 94$ the reported level is a frozen calibrated bound, not a theorem. And the unsurveyed-target deployment uses a leave-one-out center, outside Theorem 3's symmetric constructions — but not outside theory: inflating the radius by $K/(K-1)$ restores finite-sample validity at every K , distribution-free (Supplementary Material, Proposition S), and the released default ships that inflation, at a width cost under 3.5%; the frozen gates still decide on the uninflated radii, so the sealed validation carries over a fortiori. (The strictly exact split-fold alternative is infeasible at survey scale: halving the calibration set below $\lceil 1/\alpha \rceil$ units makes the conformal radius infinite.)

Efficiency. The low- ρ deconvolution-to-clustered width ratio is $1 - \frac{1}{2}\rho^2 + O(\rho^4)$, proved; the conservative band contains the clustered one pointwise; and selection cannot inflate miscov-
erage (Theorem 5). The remaining width results, one of them open, are in the Supplementary Material.

5 Simulation and a design-preserving stress test

Simulation establishes two facts: attaching the band to the wrong unit collapses coverage, and the safe selector meets its guarantees on data it has never seen.

5.1 The wrong unit collapses coverage

On a ground-truth simulation ($K = 30$ countries, $T = 6$ thresholds, errors correlated across rounds and thresholds) we ask each band for *trajectory* coverage, the whole curve-path at

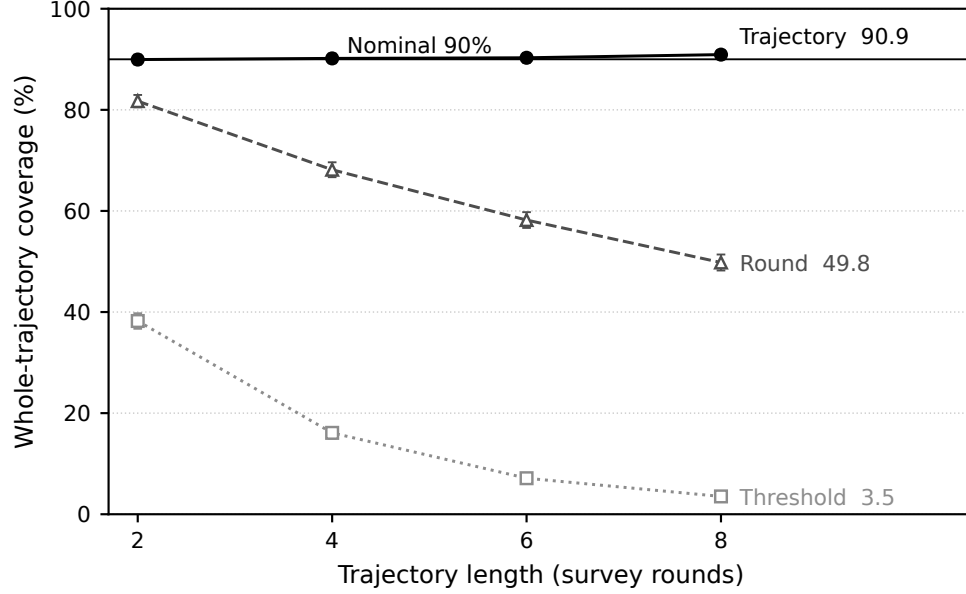


Figure 1: Coverage of a whole country trajectory, by the unit the score is attached to (4,000 replicates, nominal 90%; bars are ± 2 Monte Carlo SEs). Only the trajectory-unit band holds its level as the trajectory lengthens.

once (Figure 1). All three bands are unstudentized and finite-sample, so only the score's unit varies. A band that is marginally valid at the round level covers the trajectory 81.7% of the time at $L = 2$ and 49.8% at $L = 8$; per threshold the figures are 38.2% and 3.5%. Only the country-trajectory band holds its nominal 90% at every length, because the trajectory is a single exchangeable unit. This is the coverage face of the paper's thesis, and the same mechanism drives the certification-count collapse of Section 7. Pre-registered, reported as produced.

Severity: what each claim can detect. At ESS-like design noise, 80% power at the *net* claim needs a per-pair decline of 0.02–0.03 CDF points (realistic secular erosion); the *persistent* claim needs 0.06–0.08, crisis-scale, and its power *falls* as the trajectory lengthens. Injecting known declines into the real ESS pairs puts the thresholds at ≈ 0.033 (net) and ≈ 0.075 (persistent), with realized size 0.001–0.007 against a nominal 0.10: markedly conservative, so non-certification is weak evidence of absence at every claim. The powered claim

for secular decline is the net span, co-headlined in Section 7.

Validation. The frozen procedure was then evaluated on a sealed set of six unseen data-generating families, with the script hash recorded before first execution: all 120 cells were compatible with the target coverage (worst 0.878), with zero activations below $K = 94$, as the floor requires. The complete three-layer validation history — including a first holdout demoted to development after a scorer audit, and a disclosed generator bug — is reported in the Supplementary Material.

A design-preserving stress test, and what it does not show. Subsampling the real AmericasBarometer, the deployed selector never fires, and the pooled $\hat{\rho}$ turnover at deep subsampling coincides with ≈ 1 PSU per stratum, where the m -of- m variance estimate collapses — a failure mode of the $\hat{\rho}$ diagnostic under sparse-PSU designs, not a saturation law; the regime characterization of Section 6 rests on full-sample estimates (Supplementary Material).

6 Evidence from the ESS and the AmericasBarometer

The regime characterization. On real data the selector reduces to the clustered conformal band. Across both surveys, both estimands, and three outcomes, cross-national inference is a uniformly *low design-noise* regime: $\hat{\rho} \leq 0.22$ on the AmericasBarometer, at most 0.29 in the ESS subgroup scan, well below the $\rho_0 = 0.47$ cutoff. Every $\hat{\rho}$ we report is a downward-biased *floor*, because the PSU bootstrap resamples m -of- m without Rao–Wu–Yue rescaling (Rao et al., 1992); the rescaled rerun confirms the floors are tight — maxima 0.28 and 0.22, still severalfold below ρ_0 , zero gate activations, every certification count unchanged (Supplementary Material). The reduction is not an assumption but the selector’s own low- ρ gate at work.

Why the deconvolution is unreachable, precisely.

Proposition 2 (Finite- K feasibility of the deployed correction). *The deployed selector activates deconvolution only if a need gate $\hat{\rho}_{\text{LCB}} > \rho_0$ and a reliability gate $D \leq \tau_D$ both hold, $D = \max_t \widehat{\text{SE}}(\hat{s}_T^2)/\hat{s}_T^2$. Both are bounded by algorithmic identities of the frozen procedure. (a) $\rho = \sqrt{\bar{v}^2}/s_{\text{plug}} \in [0, 1)$ with $s_{\text{plug}}^2 = s_R^2 + \bar{v}^2$: the need gate fires only when design noise is an appreciable fraction of the between-population signal. (b) Since $\hat{s}_T \leq s_{\text{plug}}$ by construction, $D \geq \sqrt{2/(K-1)}$, so the reliability gate requires $K \geq 1 + 2/\tau_D^2$, i.e. $K \geq 94$ at the frozen $\tau_D = 0.147$. (the Supplementary Material separates the deterministic statement from its kurtosis-dependent interpretation.)*

The two requirements pull apart on real surveys (Figure 2). On the ESS the noisiest subpopulations raise $\hat{\rho}$ only to 0.29 while $K \leq 33$ leaves $D \geq 0.25$: neither gate opens (42 cells, zero activations). The WVS is the mirror image: its full item-coverage sets ($K = 95$ – 105) make the reliability gate feasible, the certification samples ($K = 59$ – 77) do not, and in either denominator the weights-only $\hat{\rho}_{\text{LCB}} \leq 0.09$ fails the need gate decisively. The survey large enough to estimate the deconvolution has almost no design noise to remove; the survey with real clustered noise has far too few countries.

Nor is the floor an artifact of our thresholds. Any *unbiased* estimator of a variance from K exchangeable populations has relative standard error at least $\sqrt{2/(K-1)}$ at the Gaussian member (Lehmann–Scheffé; Supplementary Material), so any correction relying on such an estimate at relative reliability τ — the class the deployed diagnostic lives in — needs $K \geq 1 + 2/\tau^2$: the frozen τ_D is one instantiation, 94 its intercept, $\sqrt{2/(K-1)}$ the law for this estimator class. With the proved width gain $1 - \sqrt{1 - \rho^2}$, the (K, ρ) plane splits into three regimes — correction *unnecessary*, *unlearnable*, *feasible* — and this paper’s datasets occupy the three corners (Figure 2).

Where the correction does live: small areas, on this same survey. The barrier is about the unit, and the ESS itself contains a finer one. Take the estimand this regime

is actually for, a region’s departure from its own *national* distribution, $D_g = F_g - F_{\text{nation}}$, and calibrate the transport band across regions, with the design SD of the difference from a single stratified-PSU bootstrap of the whole country (construction and exclusion rules in the Supplementary Material). On that estimand the design-to-total ratio is not marginal: $\hat{\rho}_{\text{LCB}}$ runs 0.26–0.52 and clears $\rho_0 = 0.47$ in eight of twenty-three unit-rounds. Reliability needs scale, and the two gates open together only once $K \gtrsim 230$: units of forty to sixty respondents, rounds 9 and 10, $K = 228$ –287. There the *frozen* selector, no constant retuned, activates the design-aware branch for the first time on real survey data: a band 21–27% narrower than the conservative envelope at the reported level 0.88, with certified width-gain lower bounds of 0.16–0.22 across twelve independent bootstrap streams.

The regions nest in twenty-seven countries, nine each, so the objection this paper would itself raise is that the effective unit is the country and K is really 27. Holding out whole countries provides the diagnostic: calibrating on the other twenty-six countries’ regions leaves the plug-in anchor at 0.89–0.90 against its 0.90 target in all twelve cells, within 0.015 of leave-one-region-out calibration — deleting nine country-mates changes essentially nothing, evidence against within-country dependence deflating the quantile. Coverage *conditional* on the country does not transfer (the worst held-out countries sit at 0.53–0.63; Supplementary Material); the guarantee is marginal over regions, as conformal guarantees are. In the activating cells the selected band covers 0.94–0.98 against its nominal 0.88, scored on the observed departure; the latent departure is unobservable, so we read the activation as feasibility — the frozen procedure enters its correction regime at this unit — with latent-target validity resting on the scale-family condition (A1) as everywhere in that branch. Two scope conditions travel with the result: reaching $K \gtrsim 230$ requires pooling regions coded at different NUTS levels (restricted to the fourteen countries sharing one level, the need gate still opens but $K \leq 140$ keeps the reliability gate shut), and round 11 never activates, its regional design noise being lower. The correction is thus not unreachable in principle, and not reachable across countries: it lives at the small-area scale, where Theorem 1’s remark said it would.

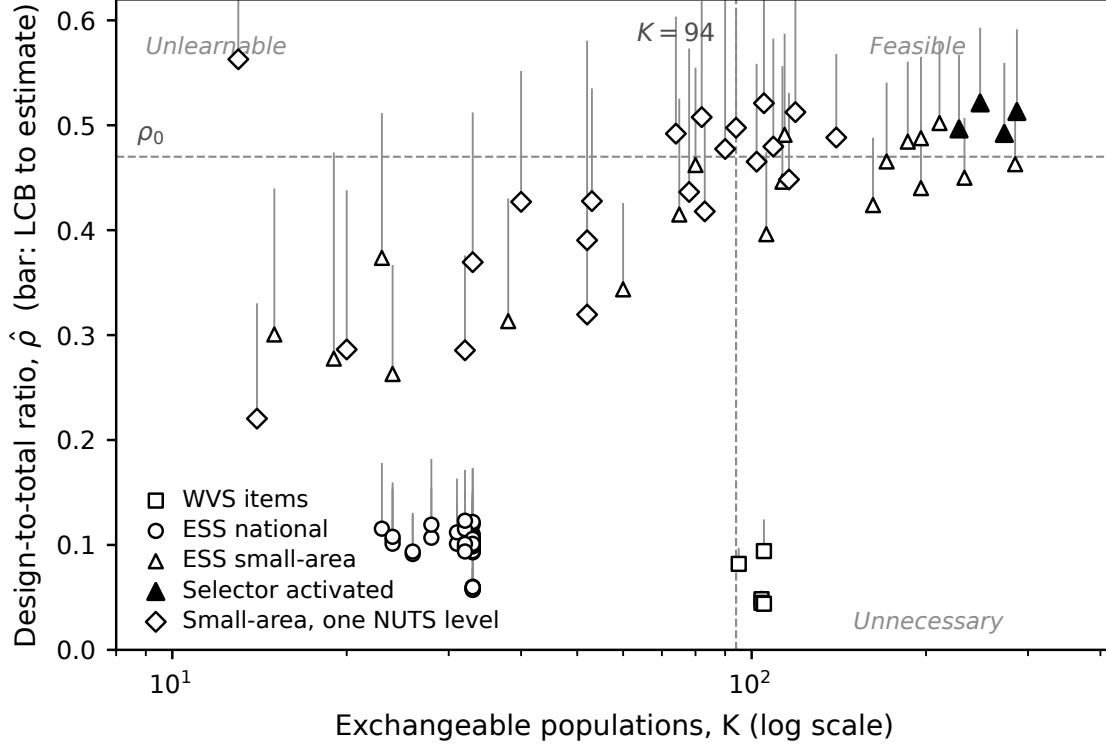


Figure 2: The feasibility frontier in the $(K, \hat{\rho})$ plane. Each marker is a real-data cell at its lower confidence bound, with a bar to the point estimate; filled markers are the four cells where the frozen selector activated. Dashed rules are the frozen boundaries, the need cutoff ρ_0 and the reliability floor $K = 94$.

How coarse can the calibration unit be? One constraint is sharp and, we believe, unremarked: the conformal level is infeasible whenever $K < \alpha^{-1} - 1$, the required order statistic exceeding the sample — a too-coarse unit is unusable, not merely inefficient; and refining is not free either, half-width falling threefold as units coarsen over the observed range (Supplementary Material). Both are statements about the *calibration* unit; an estimand’s unit is fixed by the question, and choosing among substantive areal partitions is the modifiable areal unit problem (Openshaw, 1984; Robinson, 1950), to which we add nothing.

7 Political reanalysis: how many democracies really declined?

We analyze trust in parliament (`trstprl`) and satisfaction with democracy (`stfdem`) in ESS rounds 9–11 (2018–2024, the window whose integrated files ship PSU/stratum identifiers), then extend every trajectory to the full record (rounds 1–11, 2002–2024), asking, claim by claim, how many countries survive. The comparative literature has grown skeptical of a continent-wide trust decline (Norris, 2017; Voeten, 2017; Wuttke et al., 2022); we add a formal procedure that says so and the wrong-unit diagnosis of why the over-reading arises — the dominant trend estimates (Valgarðsson et al., 2025) rest on latent-trait models with no design-based uncertainty layer (Tai et al., 2024), and their data end where our window begins. The closest competitor, latent-mood estimation (Claassen, 2019, 2020a; Kołczyńska and Bürkner, 2026), pools countries into model-based credible intervals; our bands trade pooling for model-free coverage at a stated claim, and the two are complements.

The counts collapse across claims. For `trstprl` over rounds 9–11, a marginal pairwise reading flags 20 of 30 countries; an any-pair design-aware band drops this to 12; net first-to-last decline holds in 6 (Austria, Belgium, Estonia, the United Kingdom, Greece, the Netherlands); and the *persistent, design-aware simultaneous* band over the preregistered low-trust core (the strongest claim) certifies one, Greece, which also survives Bonferroni across countries. One disclosure is essential. *Greece fielded only rounds 10 and 11 here*, so its trajectory is a single pair and the top claims coincide for it (as for Czechia and Israel, which do not certify). Among the 27 two-pair countries none certifies persistence, while the United Kingdom, Belgium, and the Netherlands pass the *plug-in* persistent claim and are demoted by the design-aware layer alone. The headline is therefore two numbers: net decline in six of thirty, and *zero* of the twenty-seven countries observed across multiple pairs certifying persistence — Greece certifies on its only available pair. The severity analysis (Section 5) explains why:

the persistent claim has 80% power only against crisis-scale declines, so its non-certifications are weak evidence of absence. For `stfdem` the pattern repeats (23 marginal, 14 any-pair, 8 net, one persistent: Greece, again one pair, failing Bonferroni there). The literature reads Greece as a crisis-era legitimacy collapse (Armingeon and Guthmann, 2014; Torcal, 2014), but those accounts concern 2009–2015 while the certified pair is 2020/21→2023/24.

The long window: the secular question, asked properly. Extending every trajectory to its full record (rounds 1–11; median six pairs, up to ten; rounds 1–8 weights-only) reframes the twenty-year narrative in both directions. Zero of the thirty-three countries with three or more usable rounds certify persistent distribution-wide decline over their full record, plug-in included. The null is not a bare one: reversing the inequality certifies *recoveries*, and across the 1,173 ordered contrasts of the trust series certified movement is directionally balanced (193 declining against 212 rising). Inside the same band, and therefore at the same α for that country, twenty-three of thirty-three countries certify at least one decline and at least one recovery, on each outcome separately and twenty on both; the two directions share one critical value by construction, so no within-country correction is needed. The count across countries is a conjunction of thirty-three coverage events and carries no familywise control by itself; the closed-testing bound below supplies it. Whatever European trust did over two decades, it was not a monotone slide (we stop short of *episodic*, which would need same-signed runs and returns toward baseline).

Under Proposition 1—one band, every ordered contrast and both directions at $\alpha = 0.10$ jointly—net erosion over the whole span certifies in eight countries (Cyprus, Spain, the United Kingdom, Greece, Hungary, Israel, Italy, Ukraine; Figure 3, Table 2). Against the per-claim construction the joint statement costs France at the span claim ($9 \rightarrow 8$) and six countries at the any-pair claim ($28 \rightarrow 22$); the persistent zero holds under both constructions (Supplementary Material).

The across-country count itself can now carry a guarantee. Inverting each joint band

Table 2: Certified net decline in parliamentary trust, 2002–2024, read off each country’s joint band (every ordered contrast, both directions, $\alpha = 0.10$). *Decline* $\geq$: simultaneous lower bound over the preregistered core, in CDF points. *Contrasts*: ordered round pairs certified declining, of all admissible; their share is the endpoint-free erosion index. The certified set is stable under weights-only design-effect inflation up to roughly 1.3; above that Spain, the marginal member, drops out (Supplementary Material).

country	rounds	decline $\geq$	contrasts	share
Ukraine	2–11	25.9	7/15	0.47
Cyprus	3–11	6.2	13/21	0.62
United Kingdom	1–11	6.1	11/55	0.20
Greece	1–11	3.1	8/15	0.53
Israel	1–11	2.6	10/28	0.36
Italy	1–11	1.6	5/15	0.33
Hungary	1–11	1.1	9/55	0.16
Spain	1–11	0.6	32/55	0.58

over its level yields a per-country certification p -value; closed testing across the thirty-three (Goeman and Solari, 2011) then certifies, with 90% simultaneous confidence, that *at least six* truly declined over their span, on each outcome, the six nameable at no further cost (Supplementary Material). The wrong-unit ladder thus closes at the top: threshold, wave, trajectory, country, count.

Under-detection shows in the ledger of contrasts: Hungary certifies its 22-year span while only 9 of its 55 ordered contrasts certify, and the United Kingdom 11 of 55, erosion that a reading built from adjacent waves alone would mostly miss. Because every endpoint choice is certified at once, the band also delivers the *share of a country’s ordered contrasts that certify a decline*: a lower bound, comparable within a country but not across (the shared critical value increases strongly with record length while the denominator grows quadratically), so we do not rank on it. Spain certifies over 32 of its 55 contrasts and Slovenia over 21 of 55, neither entering the eight because the span contrast does not certify — the endpoint sensitivity the joint band exposes. Ukraine’s certified span decline of at least 25 points runs from the post-Orange-Revolution wave of 2005 to a wartime sample: a certified statement about those two waves.

The reclassified countries are the weak-signal, higher-noise cases—*certified by the marginal plug-in but inconclusive under the design-aware band*, never “false positives,” a term we reserve for simulation with known truth.

Two confounds this window cannot exclude. Round-10 fieldwork coincided with the pandemic’s rally-round-the-flag movements, so “persistent decline through this window” is a claim made against a rally; and nine countries switched to self-completion at round 10, confounding wave-pair shifts with mode effects, a bias no design bootstrap absorbs. A mode audit built from the data’s own mode variable bounds it: the Greek pair is mode-constant and net counts are unchanged when the switchers’ confounded pairs are excluded, only their any-pair claims being affected. A preregistered age-group analysis finds no pooling artifact: youth certify zero persistent declines, an underpowered null rather than stability (Supplementary Material).

A second, global target: Foa–Mounk deconsolidation. We preregistered a reanalysis of the “democratic deconsolidation” thesis (Foa and Mounk, 2016, 2017) on the WVS/EVS trends (1981–2022, 59–77 countries with ≥ 2 waves), recoding the five-item battery so that deconsolidation is a persistent decline over each scale’s preregistered core. The WVS ships no design metadata, so its role here is a substantive stress test of the deconsolidation thesis, with the ESS carrying the design-based validity story. The verdict is calibration, not denial, but the surviving phenomenon is not the one the thesis was about. A *marginal wave-pair reading of the same data*, a criterion we construct to instantiate the bottom rung, since Foa and Mounk’s own empirics are cohort comparisons, flags 36–64 countries per item against 6–17 (8–24%) at the persistent rung. That 2.6–6.5 $\times$ ratio compounds two changes, and we separate them: holding the inference layer fixed, the rung alone accounts for 1.7–4.8 $\times$ (plug-in) or 1.9–4.8 $\times$ (design-aware), and propagating design uncertainty at fixed rung accounts for the remainder. The unit is the larger effect, but not the whole of the headline number. Three caveats. The WVS ships no PSU/stratum identifiers, so the weights-only

bootstrap understates design variance, which can only *inflate* counts: at variance $\times 1.5$ and $\times 2.0$ the certified core holds at 13 and 12 countries and the gap widens to 3.0–8.3 $\times$, so the reported gap is a lower bound. The per-item lists carry no across-country familywise control, so we aggregate to multi-item co-certification before interpreting geography. And the certification sets sit below the $K \geq 94$ bound, so every WVS band uses the full- α anchors of Theorem 5. The youth “embrace authoritarianism” claim is only half-supported (Supplementary Material).

The certified core: where the surviving deconsolidation lives. Aggregating per-item certifications yields a positive characterization (Figure 4): of 38 countries certified on ≥ 1 item, 13 form a certified core on ≥ 2 , headed by Albania on four. The core is post-communist and Arab-Spring states with fragile Latin American and African cases alongside, not the mature democracies the thesis (Foa and Mounk, 2016, 2017) concerned, formalizing with per-country error control the decline-from-euphoric-baselines pattern the post-communist literature described qualitatively (Mishler and Rose, 1997; Inglehart and Catterberg, 2003). Three qualifications travel with it. Five core members are electoral autocracies in the V-Dem classification, where falling stated support reflects autocratic legitimation rather than deconsolidation in the consolidated-democracy sense. The West enters the core exactly twice (Finland and Switzerland, on the strongman/army pair, which we flag as a probable administration artifact), with single-item entries elsewhere: no consolidated Western democracy certifies on support for a democratic system, consistent with Wuttke et al. (2022), though not that the West is absent. And the geography is partly window-contingent: recertifying on fieldwork ending by 2017 (e46) keeps eight of the thirteen but drops all three Arab-Spring cases, so the post-communist component is window-stable and the Arab-Spring component rests on WVS wave 7. Certified attitudes do not forecast institutions: core membership predicts subsequent V-Dem electoral-democracy decline no better than the marginal flag ($n = 13$ versus 54; Fisher $p = 0.89$), too small a comparison to estab-

lish the decoupling recent panel work anyway expects (Ouattara and van der Meer, 2023; van der Meer and Janssen, 2025). The estimand is the response-distribution trajectory itself: certification is descriptive error control, not a forecast. Item-level caveats are in the Supplementary Material.

8 Discussion

Uncertainty in claims about repeated cross-national surveys is usually computed for the wrong unit — twice. We gave a multiplicity-honest simultaneous band for the claim unit and a hard boundary for the calibration unit: non-identified without the design-noise law (Theorem 1), unreachable for the deployed correction at cross-national scale with it (Proposition 2). At the national unit the method reduces exactly to clustered population conformal — certifying, with the reported ρ , that the simple band is the right one; the deconvolution regime needs many exchangeable units and comparable design noise, never both across countries but reachable one unit down (Section 6). Design uncertainty, in short, does not imply that one should correct for it; sometimes the valid response is to certify that the correction is not learnable at the available unit.

Limitations. Four scope conditions, stated fully in the Supplementary Material. The machinery prices sampling error only: measurement non-invariance (wording changes, house effects, mode switches) produces exactly the shifts the bands certify, so certifications straddling such changes are conditional on measurement constancy. The deployed m -of- m PSU bootstrap (Rao et al., 1992; Wolter, 2007) biases design variance downward, though the Rao–Wu–Yue rescaled rerun leaves every count and gate unchanged. Transport-band exchangeability is a superpopulation postulate over a non-sampled, panel-attributing country set, and holds worse for the post-communist bloc. And without design metadata (WVS) only weights-only replication is possible.

The general lesson. Attaching uncertainty to the right unit changes substantive conclusions: persistence in none of thirty-three European democracies over two decades yet span erosion in eight, mostly invisible to adjacent-wave readings; a certified deconsolidation set several-fold smaller, concentrated outside the consolidated West. The pattern is not confined to surveys: wherever prediction is calibrated on estimated objects — small-area model outputs, learned features, meta-analytic effects (Remark 1) — the calibration targets carry their own error, and the claim unit is often coarser than the trend asserted. Absence of certification is not certification of absence: the severity analysis prices what each claim can detect. The right response is not to widen every band reflexively but to state the strongest object the data support, correct the estimation noise when it can be identified, and abstain when it cannot.

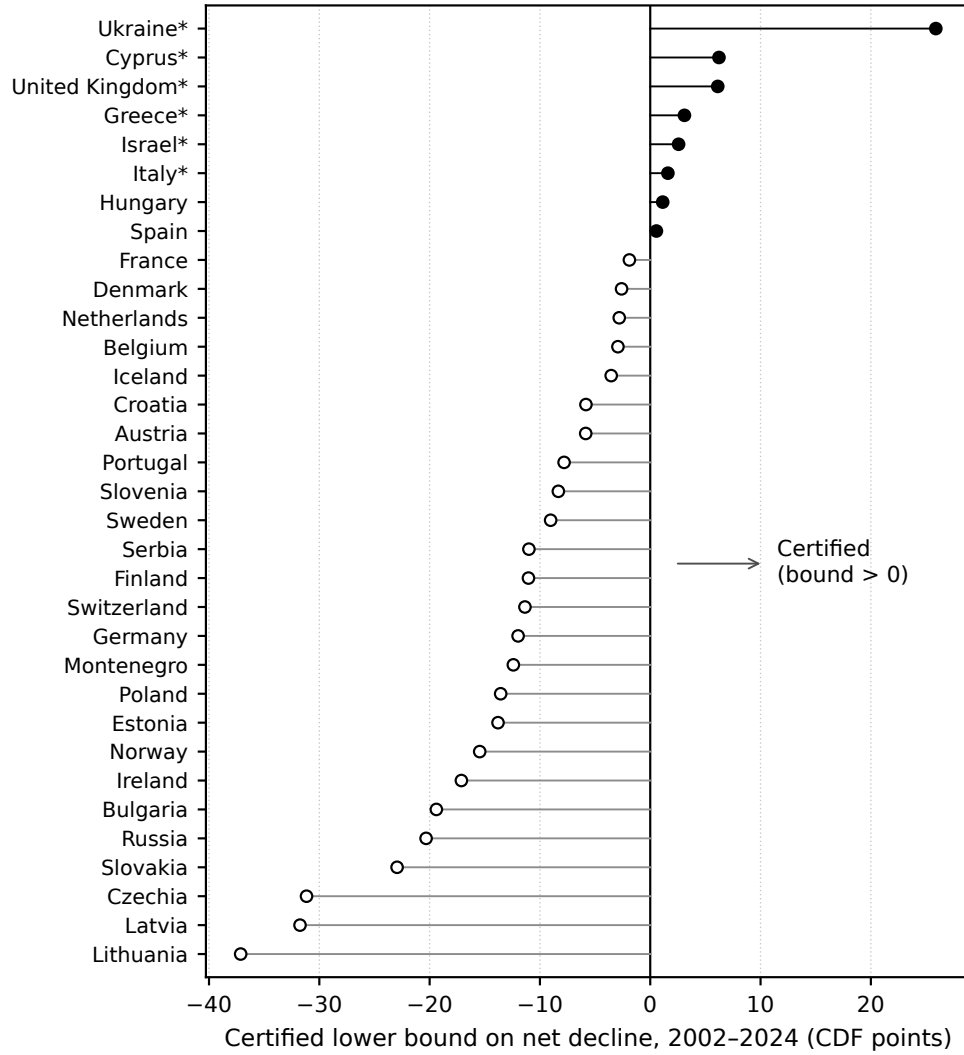


Figure 3: Each country’s simultaneous lower bound on the net (first-to-last) decline in parliamentary trust, 2002–2024, read off its joint band at $\alpha = 0.10$. Filled markers: the eight countries whose bound exceeds zero; asterisks: the six named by the closed-testing prevalence bound. Negative bounds say how far a country sits from certifying, not that it improved.

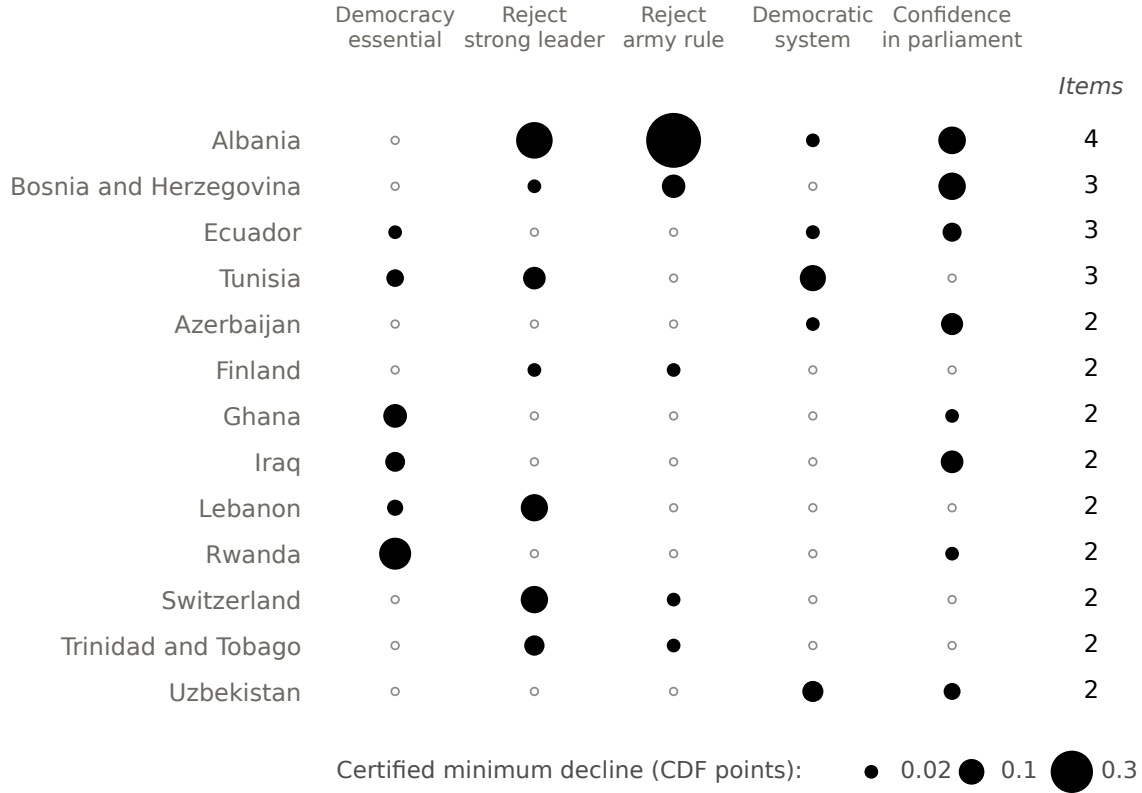


Figure 4: The thirteen-country certified core, by evidence strength (WVS/EVS 1981–2022; $K = 59\text{--}77$ per item). Circle area is the persistent band’s simultaneous lower bound on the per-pair decline, in CDF points; open circles mark items that do not certify. The full thirty-eight-country matrix is in the Supplementary Material.

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Data Availability Statement

Replication code for this article is available at <https://github.com/kunhailP/design-aware-conformal>, the package will be deposited in the Political Analysis Dataverse with a version tag upon conditional acceptance. Generative-AI disclosure (code and data analysis): substantial portions of the analysis and replication code, including the figure-generation code, were written with the assistance of Claude (Anthropic; Claude Code command-line interface, July–August 2026), under the author’s direction; all AI-assisted code is public in the repository, was reviewed by the author, and is exercised by the package’s contract-test suite and its bit-identical reproduction checks. Simulation, theory, and benchmark results are deterministic under fixed seeds with no external data, and each theorem is paired with an automated con-

tract test checking its computational implications. The survey microdata (European Social Survey ERIC, 2024; Haerpfer et al., 2024; LAPOP Lab, 2023) are licensed to their providers and cannot be redistributed; all code that runs on them is public, and the replication package documents the exact file editions, variables, retrieval steps, and sha256 checksums, alongside the public V-Dem and Claassen inputs (V-Dem Institute, 2025; Claassen, 2020b). Three headline analyses have been verified to reproduce the committed result tables bit-identically from the raw licensed files in two independent environments. The editor has been notified of the restricted access at submission.

Competing Interests

The author declares none.

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